

Lecture 4 (before summary)

27.10.2025.

Alternating series (when not all the terms are positive)

↳ terms alternate between "+" and "-".

$$\sum_{m \geq 1} x_m \text{ alt. series if } x_m \cdot x_{m+1} \leq 0$$

$$\text{Ex: } \sum_{m \geq 1} (-1)^m \cdot \underbrace{a_m}_{>0}$$

$$\sum_{m \geq 1} (-1)^{m+1} \cdot \underbrace{a_m}_{>0}$$

$$\sum_{n \geq 1} \frac{(-1)^{n+1}}{n} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \ln 2.$$

↳ Is this series convergent ($\rightarrow \ln 2$)? Yes.

Proof:

• Let $S_n = x_1 + x_2 + \dots + x_n \rightarrow$ partial sum of the series

• Take $S_{2k+2} - S_{2k} = \frac{1}{2k+1} - \frac{1}{2k+2} = \frac{1}{(2k+1)(2k+2)} > 0$

$$\Rightarrow S_{2k+2} > S_{2k}$$

$\Rightarrow$ even indexes $\rightarrow$ (1)

• Take $S_{2k+3} - S_{2k+1} = -\frac{1}{2k+1} + \frac{1}{2k+3} = -\frac{1}{(2k+1)(2k+3)} < 0$

$$\Rightarrow S_{2k+3} < S_{2k+1}$$

$\Rightarrow$ odd indexes $\rightarrow$ (2)

• $S_{2k+1} - S_{2k} = \frac{1}{2k+1} \rightarrow 0$ as $k \rightarrow \infty$. $\Rightarrow S_{2k+1} > S_{2k}$ (4).

$$\Rightarrow \lim_k (S_{2k+1} - S_{2k}) = \lim_k \frac{1}{2k+1} = \frac{1}{\infty} = 0.$$

$$\Rightarrow \lim_k S_{2k+1} = \lim_k S_{2k} \quad (3)$$

(1), (2), (3), (4) $\Rightarrow S_{2k}, S_{2k+1}$ are convergent.

$$\text{and } \lim_k S_{2k} = \lim_k S_{2k+1} = L \in \mathbb{R}.$$

$\Rightarrow$ since we have 2 subsequences and the sequence $\lim \Rightarrow$ convergent.

$$\Rightarrow S_n \text{ convergent to } L = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots$$

• For now, we have L finite number. $L = ?$
 $\Downarrow$
 conv.

Recall: $\lim_n \left(1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} - \ln n\right) = \gamma$ (GAMMA
 euler's number $\in (0, 1)$
 $\gamma = 0.577 \dots \in (0, 1)$)
 (THE HARMONIC SERIES)
 behaves like the logarithm

- Let's take one of them: $(S_{2k}$ or $S_{2k+1})$

$$S_{2m} = \sum_{k=1}^{2m} \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots + \frac{1}{2m-1} - \frac{1}{2m}$$

all positive
(harmonic)

$$1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{2m-1} + \frac{1}{2m}$$

$$- 2 \cdot \left(\frac{1}{2} + \frac{1}{4} + \dots + \frac{1}{2m} \right)$$

bring in the $\ln$

$$\left(1 + \frac{1}{2} + \dots + \frac{1}{2m} - \ln 2m \right) + \ln 2m$$

$$- \left(1 + \frac{1}{2} + \dots + \frac{1}{m} - \ln m \right) - \ln m$$

$$\lim_n \left[\underbrace{\left(1 + \frac{1}{2} + \dots + \frac{1}{2m} - \ln 2m \right)}_{\rightarrow \gamma} + \ln 2m - \underbrace{\left(1 + \frac{1}{2} + \dots + \frac{1}{m} - \ln m \right)}_{\rightarrow \gamma} - \ln m \right]$$

$$= \lim_n (\ln 2m - \ln m + \cancel{\gamma} - \cancel{\gamma}) = \lim_n \ln \frac{2m}{m} = \ln 2.$$

$$\Rightarrow \lim_n S_{2m} = \ln 2$$

$$\Rightarrow 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots \rightarrow L = \ln 2.$$

As you can see, the convergence for alternating series is not easy to prove. This is why we have:

Leibniz Test: Let $(x_n) \searrow 0$. Then $\sum_{n=1}^{\infty} (-1)^n \cdot x_n$ convergent.

Proof:

Let $x_n \searrow 0$.

Goal: analyze $\sum_{n=1}^{\infty} (-1)^n \cdot x_n$.

• Let $S_m = \sum_{k=1}^m (-1)^k \cdot x_k \rightarrow$ partial sum.

I want to show that S_n is convergent.

↳ I will show that S_n is fundamental (Cauchy) $\Leftrightarrow (S_n)$ convergent.

S_m is Cauchy if $|S_{m+p} - S_m| < \varepsilon$.

so $|S_{m+p} - S_m| \rightarrow 0$ as $m \rightarrow \infty$.

$\Rightarrow$ after an index, all terms get close to...

$$|S_{m+p} - S_m| = |(-1)^{m+1} \cdot x_{m+1} + (-1)^{m+2} \cdot x_{m+2} + \dots + (-1)^{m+p} \cdot x_{m+p}|$$

(we wrote it directly. the first m terms are the same in both S_{m+p} and S_m)

$$\begin{array}{c} \frac{|-1|=1}{\frac{x_m \geq 0}{x_{m+2} < x_{m+1}}} \quad \begin{array}{c} \xrightarrow{>0} \\ x_{m+1} + x_{m+2} + \end{array} \quad \begin{array}{c} \xrightarrow{>0} \\ x_{m+3} - x_{m+4} + \dots + x_{m+p-1} + x_{m+p} \end{array} \end{array}$$

even amount of elements
 $\Rightarrow$ everyone has a pair.

$$\Rightarrow \underbrace{1 \cdot 1}_{>0}$$

odd amount of elements
 $\Rightarrow$ not every one has a pair

$$\Rightarrow \underbrace{1 \cdot \dots \cdot 1}_{>0} \quad \underbrace{1 \cdot \dots \cdot 1}_{>0} \quad \text{always.}$$

$$\begin{array}{c} \text{get rid of } | | \\ \hline \text{because } > 0 \end{array} \quad x_{m+1} - \underbrace{x_{m+2} + x_{m+3}}_{<0} - \underbrace{x_{m+4} + \dots}_{<0}$$

$$\begin{array}{c} \frac{x_m \geq 0}{x_{m+3} - x_{m+2} < 0} > |S_{m+p} - S_m| < x_{m+1}. \end{array}$$

$$\Rightarrow 0 \leq \lim_n |S_{m+p} - S_m| \leq \lim_n x_{m+1} = 0.$$

$\Rightarrow (S_m)$ Cauchy seq.

Absolutely convergent series $\rightarrow$ we make all of the nr. to be positive.

$\sum_{n \geq 1} x_n$ is absolutely convergent if $\sum_{n \geq 1} |x_n|$ convergent.

Ex: $1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \dots = \ln 2 = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n}$ convergent.

Is it absolutely convergent? $\sum_{n=1}^{\infty} \left| \frac{(-1)^{n+1}}{n} \right| \stackrel{?}{=} \sum_{n=1}^{\infty} \frac{1}{n} = \infty$?

$\hookrightarrow$ NO.

Absolutely convergent $\Rightarrow$ convergent

~~*~~

$\sum_{n=1}^{\infty} |x_n|$ convergent.

$$\overline{S_n} = |x_1| + |x_2| + \dots + |x_n| \text{ conv. } \Rightarrow \text{Cauchy}$$

↓
partial sum.

$$|\overline{S_{m+p}} - \overline{S_m}| \stackrel{(*)}{=} |x_{m+1}| + \dots + |x_{m+p}| \rightarrow 0 \text{ as } m \rightarrow \infty.$$



$$\text{Let } S_m = x_1 + x_2 + \dots + x_m.$$

Let's show that (S_m) is also Cauchy. $\Rightarrow (S_m)$ convergent.

$$0 \leq |S_{m+p} - S_m| = |x_{m+1} + x_{m+2} + \dots + x_{m+p}| \leq |x_{m+1}| + |x_{m+2}| + \dots + |x_{m+p}| \rightarrow 0 \text{ as } m \rightarrow \infty$$

$$\Rightarrow \lim_n |S_{m+p} - S_m| = 0 \Rightarrow (S_m) \text{ is Cauchy.}$$

Theorem (Cauchy) for abs. conv.:

- take the natural num. $\Rightarrow$ reorder (= bijection) $\Rightarrow$ permutation
- $\Rightarrow$ the series of this permutation is also convergent.

(I can rearrange the elements and get the same value (sum))

Let $\sum_{n=1}^{\infty} x_n$ be an abs. conv. series

$\sigma: \mathbb{N} \rightarrow \mathbb{N}$ a bijection

Then $\sum_{n=1}^{\infty} x_{\sigma(n)}$ is abs. conv. and $\sum_{n=1}^{\infty} x_{\sigma(n)} = \sum_{n=1}^{\infty} x_n.$

„Any rearrangement of an absolutely convergent series has the same sum“.

Conditionally convergent series $\rightarrow$ the order matters

$$\sum_{n=1}^{\infty} x_n \text{ cond. conv. if: } \begin{cases} \sum_{n=1}^{\infty} x_n \text{ converges} \\ \sum_{n=1}^{\infty} |x_n| \text{ diverges.} \end{cases}$$

$$x_n = \frac{(-1)^{n+1}}{n} \begin{cases} \rightarrow 0 \text{ (conv.)} \\ || \rightarrow \infty \text{ (div.)} \end{cases}$$

Theorem (Riemann)

$\sum_{n=1}^{\infty} x_n$ cond. conv. series and $x \in \mathbb{R}$. Then $\exists \sigma: \mathbb{N} \rightarrow \mathbb{N}$ s.t.

$$\sum_{n=1}^{\infty} x_{\sigma(n)} = x.$$

$\Rightarrow$ a cond. conv. series can be rearranged to converge to any value or diverge to $\pm \infty$.

$\therefore$ We can choose anything and we will find a bijection such that $\sum_{n=1}^{\infty} x_{\sigma(n)} = x$.

Ex: Rearrange the alternating harmonic series to obtain a $\neq$ sum ($\neq \ln 2$).

$$1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \dots = \ln 2.$$

$$\begin{aligned} 1 - \frac{1}{2} - \frac{1}{4} + \frac{1}{3} - \frac{1}{6} - \frac{1}{8} + \dots &= \frac{1}{2} - \frac{1}{4} + \frac{1}{6} - \frac{1}{8} + \frac{1}{10} - \frac{1}{12} + \dots \\ &= \frac{1}{2} \left(1 - \frac{1}{2} + \frac{1}{3} - \dots \right) \\ &= \frac{1}{2} \cdot \ln 2. \end{aligned}$$

ABS. CONV.

if we rearrange the terms,
we obtain the same sum.

COND. CONV.

if we rearrange the terms,
we obtain a $\neq$ sum each time.
 $\rightarrow$ that sum can be any number?

Power series $\rightarrow$ based on the powers of x .

(a_m) , $c \in \mathbb{R}$.

$\sum_{m=0}^{\infty} a_m \cdot (x-c)^m$ is a power series centered at c .

Theorem (radius of conv.)

$\sum_{m=0}^{\infty} a_m \cdot (x-c)^m$ pw. series.

$\exists! R \in [0, \infty)$ s.t. $\sum_{m=0}^{\infty} a_m \cdot (x-c)^m$: $\left\{ \begin{array}{l} \text{converges abs. when } |x-c| < R \\ \text{diverges when } |x-c| > R \end{array} \right.$

Theorem (how to obtain R)

If $\exists \lim_n \sqrt[n]{|a_m|} = L \in [0, \infty]$, then $\sum_{m=0}^{\infty} a_m \cdot (x-c)^m$ has:

$$R = \begin{cases} \frac{1}{L}, & \text{if } L \in (0, \infty) \end{cases}$$

$$R = \begin{cases} 0, & \text{if } L = \infty \end{cases}$$

$$R = \begin{cases} \infty, & \text{if } L = 0 \end{cases}$$

! $R=0$ when $x=c$

Corollary

If $\exists \lim_n \left| \frac{a_{m+1}}{a_m} \right| = L \in [0, \infty]$, then:

$$R = \begin{cases} \frac{1}{L}, & \text{if } L \in (0, \infty) \end{cases}$$

$$R = \begin{cases} 0, & \text{if } L = \infty \end{cases}$$

$$R = \begin{cases} \infty, & \text{if } L = 0. \end{cases}$$

Proof: $\sum_{n=1}^{\infty} a_n \cdot (x-c)^n$ is abs. conv. if $\sum_{n=1}^{\infty} \underbrace{|a_n| |x-c|^n}_{x^n}$ conv.

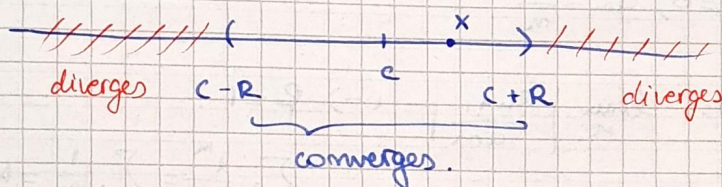
Root test: $\sqrt[n]{x_n} = x_n^{\frac{1}{n}} = |a_n|^{\frac{1}{n}} \cdot |x-c|^{\frac{1}{n}} = \sqrt[n]{|a_n|} \cdot |x-c|$

Assume $\sqrt[n]{|a_n|} \rightarrow L$.

$\Rightarrow \sqrt[n]{x_n} = L \cdot |x-c|$

$\begin{cases} < 1 \Rightarrow \sum x_n \text{ conv.} \\ > 1 \Rightarrow \sum x_n \text{ div.} \end{cases}$

$\Rightarrow L \cdot |x-c| < 1$ if $|x-c| < \frac{1}{L} = R$.



Ratio test: $\frac{x_{n+1}}{x_n} = \left| \frac{a_{n+1}}{a_n} \right| \cdot |x-c| \rightarrow L \cdot |x-c| \dots$

Convergence set: $C = \{x \in \mathbb{R} \mid \sum_{n=0}^{\infty} a_n \cdot (x-c)^n \text{ converges}\}$.

C contains $(c-R, c+R)$ and possibly the endpoints.

When $\sum_0^{\infty}$ and $\sum_1^{\infty}$?

Examples: convergence of a power series.

① $\sum_{n=0}^{\infty} x^n$, $c=0$, $a_n=1$.

Radius of convergence: $\lim_n \left| \frac{a_{n+1}}{a_n} \right| = 1 = L$.

$\Rightarrow \sum_{n=0}^{\infty} x^n$ is absolutely convergent, $(\forall) x \in (-1, 1) \rightarrow (0-R, 0+R)$

The convergence set: check for $x=1$: $\sum_{n=0}^{\infty} 1^n = \infty$?

$x=-1$: $\sum_{n=0}^{\infty} (-1)^n$ divergent

$\Rightarrow C \in (-1, 1)$

② $\sum_{n=1}^{\infty} n \cdot x^n$, $c=0$, $a_n = n$.

Radius of convergence: $\lim_n \left| \frac{a_{n+1}}{a_n} \right| = 1 = L \Rightarrow R = \frac{1}{L} = 1$.

$\Rightarrow \sum n \cdot x^n$ is abs. conv, $(\forall) x \in (-1, 1)$.

Convergence set: check for $x=1$: $\sum n = \infty$
 $x=-1$: $\sum n \cdot (-1)^n$ div. $\} \Rightarrow c \in (-1, 1)$

③ $\sum_{n=1}^{\infty} \frac{x^n}{n}$, $c=0$, $a_n = \frac{1}{n}$

R. of conv: $\lim_n \left| \frac{a_{n+1}}{a_n} \right| = 1 \Rightarrow R = 1$.

Convergence set: check for $x=1$: $\sum \frac{1}{n} = \sum \frac{1}{n} = \infty$
 $x=-1$: $\sum \frac{(-1)^n}{n} = -\ln 2$ $\} \Rightarrow c \in (-1, 1)$

Theorem $\rightarrow$ rad. of conv.

$\rightarrow$ puteum integra / deriva un power series.

"Integral of the sum = sum of the integral"

"Derivative of the sum = sum of derivative"

$\sum_{n=0}^{\infty} \frac{x^n}{n!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!} = \dots = \exp(x) = e^x$

$\exp'(x) = \exp(x)$

$\exp(0) = 1$.

$e^x = \exp(x) = \sum_{n=0}^{\infty} \frac{x^n}{n!}$

Rad. of conv: $\lim_n \left| \frac{a_{n+1}}{a_n} \right| = \lim_n \frac{1}{n+1} = 0 = L$

$\Rightarrow R = \frac{1}{0} = \infty$.

$\Rightarrow \sum_{n=0}^{\infty} \frac{x^n}{n!}$ is abs. conv., $(\forall) x \in \mathbb{R}$

$\Rightarrow$ I am always inside the interval $(c-R, c+R)$

Take the derivative: $\exp'(x) = \sum_{n=0}^{\infty} \left(\frac{x^n}{n!} \right)' = \left(1 + x + \frac{x^2}{2!} + \dots + \frac{x^n}{n!} + \dots \right)'$

$$= 1 + \frac{2x^1}{2!} + \frac{3x^2}{3!} + \dots + \frac{n! x^{n-1}}{n! (n-1)!}$$

$$= 1 + x + \frac{x^2}{2!} + \dots + \frac{x^{n-1}}{(n-1)!}$$

$$\vdots \quad \sum_{n=1}^{\infty} \frac{x^{n-1}}{(n-1)!}$$

$$\vdots \quad \sum_{n=0}^{\infty} \frac{x^n}{n!}$$

$$= \exp(x).$$